

INTRODUCTION TO DATA SCIENCE: INTEGRATION OF DATA SCIENCE, ARTIFICIAL INTELLIGENCE AND BIG DATA ANALYTICS FOR FORENSIC SCIENCE

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EXPECTED LEARNING OUTCOMES 1

- By the end of this Course Unit, learners should be able to;
 1. Define Data Science and related Terminologies.
 2. Compare and Contrast Data Science with Big Data Analytics, Data Science with Data Analysis, Artificial Intelligence with Machine Learning, Data Science with Data Mining.
 3. Outline the Pillars and Cornerstones of Data Science, Artificial Intelligence and Big Data Analytics.
 4. Describe the important roles of Data Science, Artificial Intelligence, Big Data Analytics and Data Mining in not only in Forensic Science, but also in the real-life.

EXPECTED LEARNING OUTCOMES

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5. Describe “How to Code for Data Science Using Python”.
6. Describe the various Forensic Statistics and their applications in Forensic Investigations.
7. Describe the Designing of Experimental Studies and Surveys for Collecting Forensic Science Evidences.
8. Describe the Basics of SQL language for querying Relational Databases in Data Science.
9. Discuss the Basic Concepts and Principles of Relational Databases and Warehousing in Forensic Science.
10. Describe the Various types of Data Analytics and their applications in Forensic Science.

TAKE AWAY MESSAGE

- *“Forensic Science / Epidemiology & Medical Statistics is Awash with Massive, Already Existing Data. The Problem is Not Lack of Data, but What To do with the Data and How to Leverage Big Data Analytics and Digital Revolution. This is Critical for Increased Efficiency, Performance Optimization, Better Allocation and Use of Resources, as well as Actionable, Profitable Insights to drive Innovations and Research”.*

DATA SCIENCE: BASICS

- Data Science:- A Subject that Unify or Combines Statistics, Computer Programming, Artificial Intelligence and Machine Learning, Big Data Analytics, Data Analysis/Analytics, into one.
- The purpose being to **Acquire, Process, Analyze Big Data** for actionable insights from the Data Analytics and to automate tasks, processes and workflows.
- A fairly new area of study with application in almost every sphere of our lives.
- Can be considered as one among the “**Disruptive Technologies**”, that are out there.

DISTINCTION BETWEEN DATA SCIENCE AND STATISTICS/DATA ANALYSIS

- Distinct from Statistics and Data Analysis because unlike in Statistics, one can customize or create own codes and use for Data Analysis.
- Also unlike Statistics/Data Analysis, Data Scientists can automate tasks/processes, procedures and/or workflows.
- Data Scientists can create automated Machine Learning-based Models and Algorithms.
- Also can integrate many principles together such as Data Modeling, AI, Big Data Analytics, Data Mining, Databases, Data Integration, Data Mapping, ETL, API Integration into One/Single Process to optimize performance and improve efficiency.

DATA SCIENCE: BASICS 2

- As a “**Data Scientist**”, you can not only customize your Data Analysis by writing your own codes and functions to analyze your data, but also can automate the task/process.
- **Data Scientists** are tasked with creating “**Models**” and “**Algorithms**” to “Crunch” Large Volumes of Datasets, to derive **Actionable, Profitable and Localized Insights** from the **Data Analytics** to drive **Innovations and Discoveries**.
- These Insights are critical for **Organizational Strategic Planning, Data-driven Decision Making, and Operational/Performance Efficiencies**.

DATA SCIENCE: BASICS 3

- Hence, to be a successful Data Scientist, the following Soft Skills and Technical Competencies are essential;
 1. Strong Analytical Skills and Data Intuition.
 2. Effective Communication.
 3. Basic Computer Application Packages (Ms-Excel & Ms-Access).
 4. Basic Statistics (Descriptive, Inferential Statistics, Statistical Modeling and Regression Analysis) and Probability (Types, Distributions, Bayesian Theorem/Rules/Networks & Analysis).
 5. Basic Coding/Writing Codes/Scripts.

DATA SCIENCE: BASICS 4

6. Artificial Intelligence (AI) Principles (Creating Machine Learning/ML Models and Algorithms).
7. Big Data Analytics, Distributed Computing and Big Data Analytics Frameworks Fundamentals.
8. Data Mining, Basic Principles of Querying Relational Databases using SQL and Data Warehousing.
9. Designing of Experimental Research and Surveys as well as Analysis (Statistical Principles of Hypothesis Testing).
10. Cloud Computing/Remote Computing and Virtualization.
11. Internet of Things (IOT) and Industrial Internet of Things (IIOT)/Machine-to-Machine (M2M) Communication.

THE DATA SCIENCE PILLARS

- Data Science relies heavily on the following 5 pillars;
 1. Data Acquisition/Sourcing
 2. Data Processing
 3. Data Analysis
 4. Data Surfacing/Visualization/Making the Data Available
 5. Data Storage & Archiving

THE CORNERSTONES OF DATA SCIENCE

- Likewise, the main “Cornerstones” of Data Science, include;
 1. Artificial Intelligence (AI).
 2. Big Data Analytics.
 3. Data Mining, Integration of Data Sources and Knowledge Discovery.
 4. Coding/Programming.
 5. Machine Learning-Based Models and Algorithms..
 6. Cloud Computing/Remote Resources.
 7. Internet of Things (IOT).

THE CORNERSTONES/PILLARS OF AI IN FORENSIC SCIENCE

1. Machine Learning (Supervised and Unsupervised Computer Learning).
2. Machine Learning-based Models and Algorithms.
3. Deep Learning/ Neural Networks.
4. Natural Language Processing (NLP).
5. Computer Programming/Languages e.g. Python.
6. Automation and Optimization.

THE CORNERSTONES/PILLARS OF BIG DATA ANALYTICS IN FORENSIC SCIENCE

1. Distributed Computing (Distributed File Systems, Distributed Storage, Distributed Processing).
2. Interconnection/Interconnectivity and Computer Networking.
3. The Big Data Analytics Frameworks.
4. Data Mining, Data Integration, Pipelining and Warehousing.
5. Automation of Tasks, Processes and Workflows.

INTRODUCTION:- DEFINITIONS 1

1. **Computer Software:-** Collection of instructions that enable a user to interact with a computer or perform a specific task (e.g. of Softwares include; System, Programming & Applications).
2. **Computer Application:-** A Program, Set of Codes or Computer Language designed or structured for performing specific tasks by a Computer.
3. **A Code:-** Set of instructions to a computer for a specific task OR a System of rules to convert one form of information into another form representation/output by a computer.
4. **Coding:-** Writing a Set of Instructions/Block of Codes for performing specific tasks/output/information representation by a computer.

DEFINITIONS 2

5. Types of Codes:-
 1. **Source Code/Source Coding:-** The original/initial code/list of instructions (In Group Projects). Source of a programme, that tells the program how to function.
 2. **Pre-compiled Code:-** and
 3. **Script:-** List of commands/codes block executed by a certain program (e.g. Automate or generate web pages).
6. **A Syntax:-** The Structure/Arrangement of text/words/numbers/statements to create a code.
7. **Computer Programming Language:-** A Computer Application or Software with Sets of Syntax of Rules/Instructions for writing “Codes” or Instructions to a computer.

DEFINITIONS 3

- **Artificial Intelligence (AI):-** The use of Computers, Machines Learning Processes, Algorithms and Training Input Data to simulate Human Intelligence by Performing Complex Tasks usually performed by Humans.
- The main aims here include increasing efficiencies, performances, knowledge management, solving complex problems and to minimize over dependency on Humans.

DEFINITIONS 4

- **Machine Learning Model:-** This is a simplified mathematical representation of the expected output of a training process of a data set, OR a set/block of codes, instructions and procedures to a computer to perform a certain task.
- **Computer Algorithm:-** This is a process or procedures made up of set/blocks of codes of rules and instructions to be followed by the computer in a problem-solving operations, OR These are procedures made of set of codes, running on a data set and instructions to perform a specific task.

DEFINITIONS 5

- **Cloud Computing:-** This refers to the delivery of computing services, systems, resources such as Servers, Storage, Databases, Tools/Resources, Networking Services, Applications/Softwares, Analytics and Intelligence “Remotely”, or over the internet (“The Cloud” or “Virtual Space”).
- This is aimed at offering on-demand Resources Availability, Flexible Resources, Faster Innovations such as Computing Power & Broad Network Access etc., and Economies of Scale.

DEFINITIONS 6

- **Internet of Things (IOT):-** This refers to the “**Network of Physical Objects**” (*“Things”*) that are embedded with Sensors, Softwares/Applications & Other Technologies for the purpose of connecting, collecting and exchanging Data with other devices and systems over the internet. Such as Wearable Devices, Car Trackers, Asset Tagging & Tracking, Sensors for “Track & Trace” for Real-Time Monitoring and Alerts.
- Digital Devices and Systems, Wearable and/or Portable Mobile Devices can collect data by recording, monitoring and adjusting each interactions between connected “Things”, with minimum human intervention.

DEFINITIONS 7

- **Industrial Internet of Things (IIOT):-** Also sometimes referred to as the “Fourth Wave of Industrial Revolution”, “Industrial 4.0”, or “Machine-to-Machine Communication”.
- IIOT refers to the application of IOT Technologies in the industrial settings to connect machines with other machines via sensors, systems and other devices for not only collecting data, but also sharing data over the internet with minimum human intervention.

DEFINITIONS 8

1. **Big Data:-** This refers to large volumes of useful/valuable, quality Data Sets (Yottabytes (YB) or 1 Septillion Bytes) of different varieties, consisting of both Unstructured and Structured Data that is/or can be rapidly generated/gathered and analyzed.
2. **Big Data Analytics:-** This, refers to examining large volumes of varied Data Sets for new knowledge discovery and to derive meaningful, actionable and profitable Insights from the analytics such as, identify trends, hidden patterns, previously unknown correlation, association or relationships, Preferences, driving factors or to predict future outcomes, etc.

DEFINITIONS 9

1. **Machine Learning:-** The use of Computers (algorithms and models), training input/output data and processes), to access large volumes of data sets/Big Data (Data mining), analyze the data, learn for themselves (from the training data input) and be able to analyze new data set and make actionable insights from the analytics
- The Insights include such as identifying patterns, predicting future outcomes, classifying data sets or clustering data into natural groupings.

ARTIFICIAL INTELLIGENCE IN FORENSIC SCIENCE 1

- **Artificial Intelligence:-** This refers to the use of computers, machines, Models and Algorithms and processes to “Simulate” Human Intelligence such as performing complex automated tasks.
- **AI** is an “umbrella term” covering a variety of distinct, but interrelated processes including;
 1. Machine Learning (ML)
 2. Deep Learning

ARTIFICIAL INTELLIGENCE IN FORENSIC SCIENCE 2

3. Neural Networks such as Neural/Natural Language Processing (NLP).
4. Robotic Process Automation (RPA).
 - The AI uses Computers and Machine Processes to simulate “Human Intelligence” and thus perform complex automated tasks.
 - Such tasks may include Identifying patterns, trends or anomalies, Understanding & Interpreting Human Languages e.g. Sounds/Voice/Speech Recognition, Images Interpretation e.g. Colour Bands in Ink Analysis, Finger Prints Analysis, etc.

MACHINE LEARNING(ML)

- Machine Learning entails training Computers Models/Algorithms using Data sets such as Historical Crime Data to Create Models able to Predict Future Outcomes, Categorize/Cluster Information/Data based on similarities e.g. **“Natural Groupings”** or **“Behaviors”** and/or to Classify Data sets into different Classes.
- ML-driven Models and Algorithms are the foundation of Data-driven decision making and automation.
- Some of the examples of ML include, Predictive, Classification, Clustering Models, Content-based Collaborative Filtering and the Modern Recommendation Systems.

DEEP LEARNING 1

- Deep Learning on the other hand, is a subset of ML that however involves larger volumes of data, more training times and more layers of ML algorithms to produce “**Neural Networks**”, that are capable of more complex tasks.
- The Neural Networks, are inspired by the structure of the “**Human Brain**” (Networks of Brain Cells & Nerves), thus consists of interconnected “**Nodes**” that process and

DEEP LEARNING 2

- Requires relatively more data for training, duration/time for the training and more layers of Algorithms, than the others.
- Hence, not only more complex, but are also capable of solving more intricate and complex problems than the others such as use of “**Smart Doors**” with sensors for human movements, Image Processing and Analysis, etc.
- Some of the common frameworks for creating and refining Neural Networks are **PyTorch** and **TensorFlow**.

DEEP LEARNING 3

- For instance, the integration of CCTV Cameras with Facial Recognition Softwares and the use of Automated Facial Recognition Systems (AFRS) Algorithms in Forensic Investigation are all examples of Deep Learning.

THE NATURAL LANGUAGE PROCESSING (NLP) 1

- The NLP or Conversational AI entails the use of ML-Driven Models and Algorithms to not just understand, but also Interpret Human Language.
- The Human Language may be Verbal/Spoken/Voice or Written Language.
- The NLP Models also interpret Sounds such of a Song.

THE NATURAL LANGUAGE PROCESSING

2

- The Natural Language Processing includes both Conversational and Generative AI.
- The Conversational Language Models includes applications such as ChatGPT, Voice Recognition, Chatbots, Zuri, and Personal Assistants.
- The Generative AI includes the Blackbox.AI, Video Generating, Tutorial Videos Making applications, etc.

THE NATURAL LANGUAGE PROCESSING 3

- Hence, are used to Summarize Texts Explain Complex Concepts, Interpreter Documents, Notes, Reports, Transcriptions & Decoding of Text/Voices/Speeches, Explain & Generate Codes, Qualitative Research Analysis, Publish Research, Speech/Voice Recognition (Dictaphone).
- For instance, Google Voice Recognition, Sound/Song Recognition, the use of “My Voice as My Password” from Safaricom.
- Other Examples of Conversational AI include the Google Assistant/Hey Google, Google Voice Matching, Digital Personal Assistants (Alex, Cortana).

THE ROBOTIC PROCESS AUTOMATION

- The RPA refers to the use of AI (ML Algorithms) in Computer Programmes to automate Administrative Task, Processes and Operational Workflows.
- Includes also the Automation of Robotic Equipment and Machines such as Fitting Drones, Human Robots, Dog Robots Unmanned Equipment/Vehicles with Automation Softwares, for Better Performances such as Border Patrols, Real-Time Data Streaming such as Real-Time Monitoring and Alerts, etc.

ARTIFICIAL INTELLIGENCE IN A NUTSHELL

- Fundamentally, All the hitherto mentioned AI processes entail the Training of Computer Systems using Input (and Output in some cases) Data, to do tasks that are usually performed by human beings.
- Such as the use of Sensors, Track and Trace, Robotics in Prisons, Border Patrols, Integrated Biometrics/Security Barriers, Drones to Track Criminals/Bandits/Rustlers, etc., Face or Voice or Sound Recognition as password/identification or interpretation of written language.

APPLICATIONS OF DATA SCIENCE IN REAL-LIFE FORENSIC SCIENCE SITUATIONS 1

- The roles of Data Science in Forensic Science can be broadly categorized into three, including:-
 1. **Deterministic Role:-** Entails Innovations and Creativity to develop new Methods, Approaches, Processes, Techniques, Strategies, Techniques, etc., for not only improving operational efficiency, but also optimizing operations or performances, such as tasks/processes/workflow atomization, new crime diagnostics, methods, to.

APPLICATIONS OF DATA SCIENCE IN REAL-LIFE FORENSIC SCIENCE SITUATIONS 2

2. **Predictive Role:-** Forecasting Future Trends or Outcomes, or Proactive anticipation of outcomes/responses to avoid disasters/interruptions in the future such as business/operational disruptions e.g. Predicting of Future Crimes such as Cyberattacks (e.g. DDoS) and/or Cyberspace as the New Warfront.
3. **Cognitive Role:-** This is more about Knowledge Management and Computer Simulations of Human Beings, or the use of Computers and Robots, Robotic Equipment to perform tasks, usually performed by Human Beings.
- The main aim here is not only simplifying repetitive Tasks and improving performances, but also to reduce human dependency.

THE DETERMINISTIC ROLE OF DATA SCIENCE IN FORENSIC SCIENCE

Some of the Deterministic Roles of Data Science include:-

1. Analytical/Problem-solving (e.g. Flagging of suspicious transactions in Real-time, Cyber-attacks, Bullying, Cyber-security, Data Privacy & Protection, Oversighting and strengthening of regulation/Control e.g. Mobile/Digital Finance Markets, Authentication, Impersonation etc.). Use Design thinking approach to problem solving (transforming a stimuli into an idea and from the idea a final product/solution).
2. Centralization of Sensitive Data (e.g. “Huduma” Number/Digital Identity Cards, CODIS, etc.) and of Integration of Data from Multiple Sources
3. Data Mining for new Knowledge, Creation of Databases, Querying of Specific Data/Information of Interest and sharing of Information (e.g. Criminal Justice System with Digital Forensics or Bioinformatics/Computational Biology).

THE PREDICTIVE ROLES OF DATA SCIENCE IN FORENSIC SCIENCE

- Entails Proactively anticipating outcomes/responses to avoid business/operational disruptions. Hence, some of the roles include:-
 1. Machine Learning Modeling for predicting future Crime Trends, Hot spots, and nature, Clustering or Classification-based analysis of Forensic Data/Information for Profiling Crime and/or Criminals' (behavior or characteristics).
 2. Determination of the Current Crime Dynamics e.g. Sophistication such as New War Fronts, Economic Espionage, Blackmail, Piracy, Terrorism, Violent Extremism, Fraud, Corruption, Money laundering, Organized Crimes.
 3. Forecasting the Evolution of Crime in the Future such as Digital Crimes, Organized Crimes, Cyber-Crime, Cyberattacks, Virtual Sexual Assaults, Cyber Bullying, Drug Smuggling, Assaults, Arson, Loan Defaulting, Tax Evasion, Court absconding, Bail Jumping, Prison break/escape, etc.

COGNITIVE ROLES OF DATA SCIENCE IN FORENSIC SCIENCE 1

- Mainly for managing knowledge to reduce human overdependency on humans, solve complex problems, automation of tasks/processes/procedures or work flows. Therefore, some of these roles include:-
 1. ML-Based Models/Algorithms for Automation of repetitive Tasks/Activities/Processes/Responses for operational efficiency and optimizing organizational performance.
 2. Deep Learning Models to understand and interpret Images/Pictures/Photographs/Signatures such as Integration of Facial Recognitions Software/Systems into CTTVs, AFRS Algorithms, etc.

COGNITIVE ROLES OF DATA SCIENCE IN FORENSIC SCIENCE 2

1. Use of Neural Language Processing Models such as Conversational Language Models e.g. ChatGPT, Generative AI e.g. Blackbox.AI, Chatbots, Dictaphone, to understand Interpret Human Languages (Spoken & Written), Summarize Text, Drafting Documents, Explain complex concepts, explaining and generating codes.
2. The use of Neural Networks such as “**Smart Doors**”, to access Security Barriers/facility as an enhanced security measures.

COGNITIVE ROLES OF DATA SCIENCE IN FORENSIC SCIENCE 3

3. Artificial Intelligence (AI) and Big Data Analytics such as use of, GPS/GIS, IP address, Google Location Services for real-time tracing, tracking and locating Criminals, use of Sensors for Track-and-Trace for Monitoring Prisoners Remotely and Real-Time Alerts on location/whereabouts of criminals.
4. Use of Cloud Computing to provide “Virtual Courtrooms”.
5. Use of Sensors for “Track and Trace” such as the Vehicles/Car Tracking Systems and Assets Tagging and Tracking for real time monitoring and Alerts.

COGNITIVE ROLES OF DATA SCIENCE IN FORENSIC SCIENCE 4

5. Robotic Processes Automation such as the use of Human/Animal Robots to perform tasks usually done by the Humans e.g. use of Robotics for Taking Roll Calls, delivering food, Border Patrols and Integrated Drones to track Criminals Terrorists, etc.
6. The use of IOT, IIOT, Cloud Computing to connect physical objects “things” to the internet and data collecting systems such as Digital Devices, Wearable Devices e.g. Smart Watches, Centralization of Sensitive Data and Automation of Data Collection, Inventory Control, Logistics Processing & Analysis, Equipment Automation, machine-to-machine communication, etc.

COGNITIVE ROLES OF DATA SCIENCE IN FORENSIC SCIENCE 5

7. Tracking Cyber-Crime:- Frauds, Cyber criminal behavior, Characteristics, Sites visited, time spent, response or output.
8. Tracking of online crime, inciters, hate speeches, organized crime, political opinions on Social Media platforms or Cyber bullying.
9. Tracking of Digital/Online markets:- Customer behavior, characteristics, interests, time spent, response or output.
10. Shifting/Filtering Spams from Emails, Piles of Data e.g. Shortlisting of Candidates for Interviews from Massive Job Application CVs/Resumes.

APPLICATIONS OF BIG DATA ANALYTICS IN FORENSIC SCIENCE 1

- Significantly, Data is the engine that drives not only artificial intelligence, but also the Big Data Analytics.
- This is because ML-based Models and Algorithms for prediction/classification/clustering are solely based on Already Existing/Historical/Archive Data.
- The Models must be trained using input data, validated using validation data set, tested using testing data set, and in some cases such as the supervised learning of Prediction and Classification Modeling, with the output data.

APPLICATIONS OF BIG DATA ANALYTICS IN FORENSIC SCIENCE 2

- Hence, the availability of already Existing/Historical Data, or Collecting adequate new data for use in building and training of the AI Models and Algorithms to carry out various tasks geared to developing credible AI solutions to local challenges/problems/questions, is critical.
- Consequently, some of the uses of Historical/Archived or Newly collected Data such as Historical or Generated Crime Data includes:-
 1. Clustering of crime hot spots, dynamics or types and then Customizing response/intervention needs, products and/or resources for stakeholders.

APPLICATIONS OF BIG DATA ANALYTICS IN FORENSIC SCIENCE 3

2. Mapping out of Crime hotspots, demographics, behavior and characteristics of criminals (Psychology of criminals), determinants/driving factors or risk factors.
3. Predicting future crime trends, nature and evolution (e.g. the emerging online-based crime of blackmail, virtual sex, soliciting for sex from minors, economic espionage, cyber bullying etc.).

APPLICATION OF BIG DATA ANALYTICS IN FORENSIC SCIENCE

4. Enable digital inclusion through technology, innovation and technology.
5. Through an integrated approach Improve/Enhance crime management operations, criminal justice system, rehabilitative and preventive interventions, ease /smoothen crime management services and delivery.
6. Designing, evaluating and implementing as well as diffusing new solutions to the chronic crime and inefficient criminal justice system challenges using big data analytics.
7. Overall to drastically reduce crime rates, enhance security and boost economic growth and development.

INTEGRATION OF BIG DATA ANALYTICS WITH DATA SCIENCE FOR FORENSIC SCIENCE

- This constitutes “**Big Data**”, “**Big Data Ecosystem**”, “**Big Data Analytics**” and “**Big Data Source**” in Forensic Science.
- The phrase “**Big Data Analytics**” was coined to refer to Data Sets that were “soo” Large ($>$ Gigabyte, exceeding the capacities), such that the “Traditional” Data Processing Soft wares simply could not manage them.
- Thus, the “**Big Data**” required new, more robust computer softwares for processing. This could be best handles by data processing frameworks.
- This inspired the development the “**Big Data Analytics Frameworks**”.

THE BIG DATA ECOSYSTEM 1

- **The Big Data Ecosystem:-** This refers to the Functional Components of computing, its Systemic Platforms and General Infrastructure an Organization uses for Collecting, Storing, Analyzing and Leveraging Big Data.
- It also includes their Enabling Tools and The Potentials of Big Data Analytics in General.

THE BIG DATA ECOSYSTEM 2

- The Functional Components and Tools for the Big Data Ecosystem, include;
 1. Computer Programming Languages
 2. Computer Applications & Packages.
 3. Models & Algorithms.
 4. Internet of Things, IIOT (Big Data Sources).
 5. Cloud Computing (Storage, Resources & Data Sources).

THE BIG DATA ECOSYSTEM 3

- The Systemic Platforms in a Big Data Ecosystem include elements such as;
 1. The Operating Systems, Networking and Filing Systems in place.
 2. The Frameworks for Big Data Analytics.
 3. The General Infrastructure (e.g. Data Mining, Data Visualization Tools, Databases, Warehousing, Data Lake, Data Pipelining, Data Mapping, ETL, API Integration, etc., a Data-driven Organization uses to Leverage Big Data Analytics.

THE BIG DATA ECOSYSTEM 4

- Thus, the Functional Component of the Big Data Ecosystem has 3 Main Elements;
 1. The Data Sources (Internet, Clouds, Digital Devices, IOT/IIOT Systems, Websites, Databases, etc.), Data Mining Methods and Tools.
 2. The Data Management (Data Integration, Data Pipelining, Data Mapping, Extraction Transformation & Loading, Storage and Processing) Component.
 3. The Data Analytic Component that includes the Data Processing and Analysis.

The Data Management Component: Data Processing and Analysis 1

- The Data Processing and Analysis falls under the Data Management Component and may include the Extraction Transform and Load function, Data Pipelining which in addition to automating the process of data processing and analysis eliminates errors.
- Other Data processing activities may include Data Structuring, Cleaning, Transforming Data, Data Aggregation, Data Integration and Data Mapping, etc.

The Data Management Component: Data Processing and Analysis

- Data Integration is whereby Data from Multiple Sources are Combined together for a single analysis.
- Data Mapping in ML is the process of connecting a data field from one source to a data field in another source to standardize your data, using manual, semi-automated and full automated mapping techniques.

The Data Management Component: Data Storage

- The Data Storage component entails using various methods and tools to store the generated, gathered and/or analyzed data.
- The Storage Methods include Storage of Data in On-Premises, Open Sources using Physical Servers and/or Storing Data in the “Cloud”.
- Other Data Storage Methods include, Warehousing and Data Lake Storage different Types of Big Data.
- The Tools for Data Storage include Databases, Languages for Querying Databases such as SQL, etc.

The Big Data Sources Component 1

- This is the Data Collection Component in the Big Data Ecosystem.
- Entails, but not limited to the following;
 1. Data mining of historical or archived data from Databases, Websites, the Internet, Social Media Platforms, etc., using various computer programming languages, Softwares, Applications, Platforms, Functions, Algorithms, etc.

The Big Data Sources Component 2

2. Data Extraction and Acquisition of Meta Data form other Studies.
3. Querying of Relational Databases.
4. Web Scrapping for Real-Time Data from the Internet or Websites.
5. Importing or Getting Data from Files or Other Sources.

The Data Storage Component: Computer Data Storage Units 1

- A Computer stores data in “**Binary**” form of “0” and/or “1”.
- When a “**Binary Digit**” is stored or presented as a “String”, it is referred to as a “ **Binary Code**”.
- The smallest fundamental size of data stored by a computer is the “Bit”, which is a binary digit of “0” and “1”.
- When 8 “Bits” (8 0s and/or 1s) are combined together they form 1 “Byte”.

Computer Data Storage Units 2

1. Byte or 8 Bits of “0s” and/or “1s”.
2. Kilobyte (KB) or 1000 Bytes
3. Megabyte (MB) or 1 Million Bytes
4. Gigabyte(GB) (9 Zeros) or 1 Billion Bytes. 1 GB is the Standard by which storage capacities are judged. In 1999, a 1 GB was considered as “Big Data”.
5. Terabyte(TB) (12 Zeros) or 1 Trillion Bytes
6. Pentabyte (PB)(20 Zeros) or 1 Quadrillion Bytes, e.g. all users’ Photos on Facebook, Google Search Engine Processes 20 PB of Data, Daily.

Computer Data Storage Units 3

7. Exabyte (EB)(24 Zeros) or 1 Quintillion Bytes
8. Zettabyte(ZB) (28 Zeros) or 1 Sextillion Bytes
9. Yottabyte (YB) (32 Zeros) or 1 Septillion Bytes, Which Today is Considered as “Big Data”

THE BIG DATA ANALYTICS: FRAMEWORKS 1

- The Big Data Analytics is a complex processes given the volume of data sets (Ybs of Data) involved.
- A common practice in Big Data Analytics is to break down a Complex Task (e.g. Task A) into a number of smaller Tasks (e.g. Tasks B, C & D) that are shared and executed by Three Computers that are connected and Networked together.
- This technique is referred to as “**Distributed Computing**”.

THE BIG DATA ANALYTICS FRAMEWORKS 2

- Infact, the “**Big Data Analytics**”, is a term coined to refer to Large volumes of Data sets, so large that the Traditional Data Processer Softwares and Applications, simply cannot manage them.
- Hence, requires more Robust Computer Softwares designed specifically for that such as Distributed Computing System, Distributed File Systems, Distributed Storage, Distributed Data Analysis, Distributed Graph Processing, etc.
- Consequently, Big Data Analytics are best handled by what are referred to as “**The Big Data Analytics Frameworks**”.

THE BIG DATA ANALYTICS FRAMEWORKS 3

- Therefore, the Big Data Analytics Frameworks are Instruments specifically designed to simplify the Processing and Analysis of “Big Data”, thus, able to handle/process and analyze Big Data Rapidly, Effectively & Securely.
- They are designed to help Businesses, Organizations to not just leverage Big Data, but also integrate Business with Big Data Practice Successfully.
- Most of the Common Frameworks (e.g. Hadoop, Apache Spark, Storm, etc.) and Quite Often they are Open-Sources, with options to purchase Support, if necessary.

THE BIG DATA ANALYTICS FRAMEWORKS 4

- Some of the Examples of The Big Data Analytics Frameworks, include;
 1. The Apache-Hadoop, an Open-Source Framework designed with a Hadoop Distributed File System and Map-reduce architecture for storing & processing Large amounts of datasets.
- The Map-Reduce architecture is a technique of splitting the data sets/files into large blocks/clusters of data.

THE BIG DATA ANALYTICS FRAMEWORKS 5

- The split files are then distributed across the Nodes in a Cluster.
- Thereafter, it transfers Codes into the Nodes for processing the Data sets in parallel.
- Thus, can be used for Distributed Storage and processing of Large Data Sets.
- The concept of “**Data Locality**” ensures that Tasks are performed on the “**Node**” that stores the data.
- Hence, it is appropriately used to process batch-related tasks or Batch streaming Data/Information.

THE BIG DATA ANALYTICS FRAMEWORKS 6

2. The Apache Spark Distributed Systems, is an Open-Source Cluster-Computing Framework or a general engine for large scale data processing designed to provide high level API in Python and Java.
 - The capability of Apache Spark for both Batch and Stream Processing makes it a “**Hybrid Big Data Analytics Framework**”

THE BIG DATA ANALYTICS FRAMEWORKS 7

- AS relies on a Data Structure known as “**Resilient Distributed Data Set**” (RDD), whereby Read-only multiple sets of data items are distributed over the entire Cluster Machines.
- This makes it an Ideal for Machine Learning as it can run on a single Machine with one Executor for every CPU Core.
- Therefore, it used to process Real-Time Streaming Data and Data in Mini-Batches, as it allows the same set of application code created for Batch Analysis to be also used for Streaming Analytics.

THE BIG DATA ANALYTICS FRAMEWORKS 8

3. The Apache Storm, is another Open-Source Circulated Ongoing Calculation Framework, whose applications/operations are planned and designed as a “**Topology**” (Physical & Logical arrangement of the Nodes) with Coordinated/Directed Acyclic Graphs (DAG Shape).
- Provides Distributed Real-Time Stream Processing of Data.

THE BIG DATA ANALYTICS FRAMEWORKS 9

- Although it is mostly written in Clojure, Apache Storm is compatible with any other Programming Language.
- The idea behind Storm is to define small, discrete operations and then compose them into Topology, which acts as a Pipeline to transform the data.
- It uses the “**Sprouts**” and “**Bolts**” to define the sources of the Data Streams and the Processing Aspects applied to the data streams.
- This allows Batch, Distributed Processing of Streaming Data in Real Time.

TYPES OF BIG DATA ANALYTICS

1. Descriptive/Diagnostic Analytics for Actionable Hind sights or to answer the Question “What Happened?
2. Predictive Analytics for Foresights or to answer the Question “What Might Happen in the Future? Or “What Might Have Happened in the Past?
3. Prescriptive Analytics for Actionable Insights, or to answer the Question “Now that it has Happened, What is Best Course of Actions to Take?

OTHER COMPONENTS/ELEMENTS OF A BIG DATA ANALYTIC SYSTEM

- A Big Data Analytic System or Platform is made up of 3 major components, including;
 1. A Real-Time Query.
 2. A SQL-based Calculating Component.
 3. A Multi-dimensional Query Component.

The Big Data: The Characteristics Features 1

- The Big Data is characterized by the Five V's of;
 1. Volume:- The sheer size of the data or the magnitude or quantity of the generated data.
 2. Value:- The usefulness or the utility of the data.
 3. Velocity:- The speed at which the data is collected or how fast data can be generated or gathered and analyzed.

The Big Data: The Characteristics Features 2

4. Variety:- The types of the Data, or The number of Points of Reference used to collect the Data.
5. Veracity:- The Quality, Validity, Reliability, Accuracy and Consistency of Data.

Applications of Big Data Analytics in Forensic Science 1

1. Business Intelligence for critical information for data-driven enterprises, for having a competitive edge.
2. Knowledge Discovery from RDBS, such as subgroups, new crime trends, hot spots, actionable and profitable Insights.
3. Organizational Strategy for data-driven Decision Making and Planning.
4. Performance Tracking and Operations Optimization.

The Sources of Big Data For Forensic Science 1

- Some of the major sources of Big Data relevant to Forensic Science, include;
 1. Transaction Processing Systems.
 2. Open-Sources(Public/Government Departments/Ministries
 3. Archived/Historical Crime Data
 4. Customers'/Clients Databases e.g. Mobile Telecommunication Services Providers.
 5. Corporates' and Organizations' Websites
 6. Relational and Other Databases (e.g. Research Databases, DNA, Genomic Sequences Databases, Index Databases e.g. CODIS, etc.

The Sources of the Big Data For Forensic Science 2

6. The Social Media Platforms.
7. E-Data, Telemetry, Electronic Records & Digital Devices e.g. Mobile Applications, Wearable Devices.
8. Internet Click Stream Logs, Real-Time Alerts/Streaming.
9. National and International Organizations e.g. MOH-K, KNBS, CDC, IRCA, WHO, etc.
10. Documents/Text/Literature/Records Screening or Review, Emails, etc.
11. Internet Click Stream Logs

The Sources of the Big Data 3

- 11. Emails and Chat Messages
- 12. Medical and Hospital Records
- 13. Connected IOT Devices/Items e.g. Sensors, Physical Objects and Systems.
- 14. Connected Industrial Internet of Things (Machine to Machine Networks and Digital Systems.
- 15. Connected Items/Sensors Data Management Systems.

The Big Data Analytics Lifecycle

- The Big Data Lifecycle consists of four distinct phases, including;
 1. The Collection of Data.
 2. The Storage of Data.
 3. The Organization, Processing and Analysis of the Data.
 4. Knowledge Creation or Discovery e.g. Sub-Groups, Deriving Actionable Insights from the Analytics.

The Data Science, AI and BIG Data Analytic Tools 1

1. **The Programming Languages** used in Big Data include Python, Ruby (R), SAS, SQL, VBA, etc.
2. **The Data Mining Tools include;** Python Libraries (Numpy, Panda, Web Scrapping, Mapping Libraries, PowerPivot, SQL, Algorithms.
3. **The Analytical Tools Include;** Python, SAS, Ruby, Ms-Excel, Google Sheets & Google Analytics, SPSS, STATA, Modules e.g. Panda, Stats.Module, JSON, Scikit-learn, XGBoost, PyTorch & TensorFlow (Deep Learning Frameworks) etc.
4. **The Visualization Tools include;** Tableau, Power-BI, Jupyter Studio, Microsoft Visio, Ms-Excel Functions (Pivot Tables, Charts, etc.),Google Sheets, Google Charts, Google Data Studio, Modules e.g. Matplotlib, etc.

The Big Data Analytic Tools 2

4. **The Relational Databases Querying Tools** include; SQL, VBA, Ms-Excel Functions, Algorithms.
5. **The Data Storage Tools** include; Relational Databases, e.g. Ms-Access, Oracle, Postgress, SQL Server, Google's Bigtable, etc.
6. **The Reporting Tools** include; Web-based Applications e.g. Flask Basics, HTML, Google Analytics, Tableau, Relational Databases e.g. Ms-Access, Oracle, SQL Server, Postgress, etc.

The Challenges and Emerging Issues in Big Data Analytics 1

1. Lack of Knowledge of the Massive Data that already exists, How to Leverage Big Data Analytics and its Massive Potentials.
2. Shortage of Competent Professionals to perform Big Data Analytics and Interpret the Results.
3. Lack of Understanding of the Big Data Analytics Methods and Tools and their Selection.
4. Data Growth Issues, such as Regulation, Control, Legislations and Standardizations.

The Challenges and Emerging Issues in Big Data Analytics 2

5. Security of the Data such as Personal Information.
6. Integration of Data from the Diverse and Multiple Sources (Methods, Tools and Expertise) for Leveraging Big Data.
7. Ethical issues of Collecting, Analyzing, Reporting, Sharing and Securing Data e.g. Personal Data, Spam Bots Menace, Integrity, accurate, honesty reporting.
8. The Emerging Issue of Repurposing, Monetizing of Data, etc.



● THANK YOU